Joint Universities Accelerator School

JUAS 2026

16. – 23. February 2026

Normal-conducting accelerator magnets
Case study: Analytical & numerical design

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CERN

Introduction

- The goal is to practice elements learned during the lectures
- Students will design, specify and optimize a real magnet
- Sample case: Bending magnet for a MedAustron beam transfer line
- Work in groups of 5-6 students during 2 half-days
- At the end, students are expected to deliver a written magnet design report

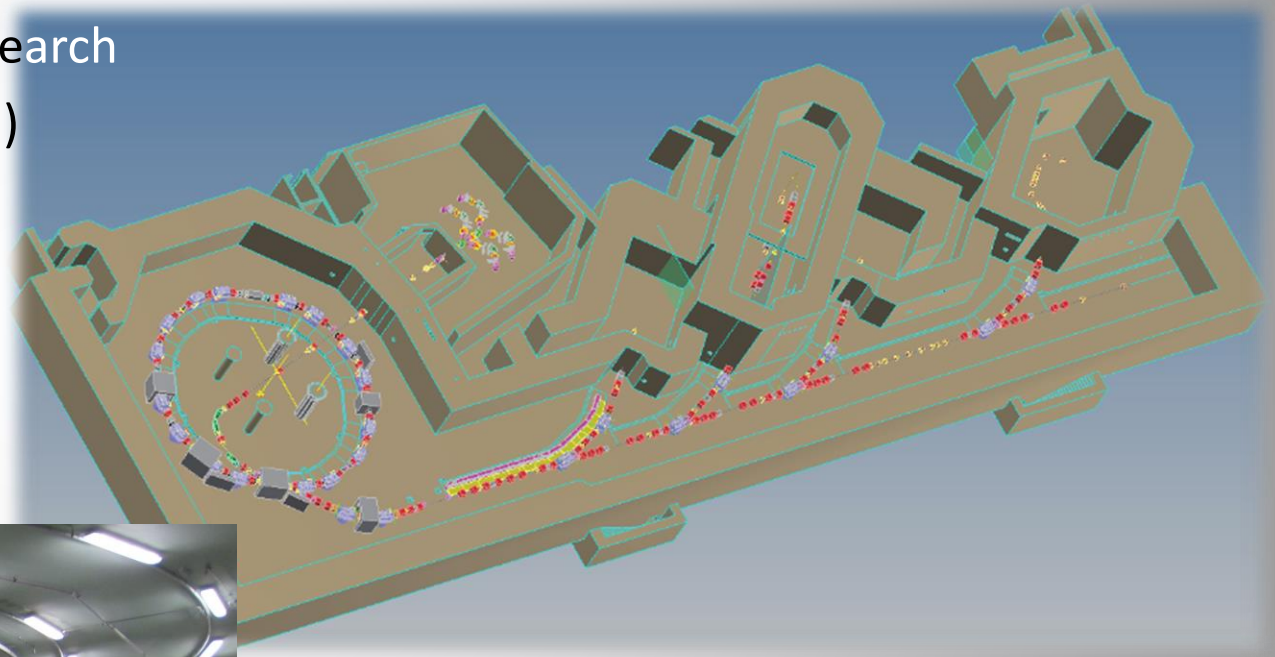
Programme

- Short introduction to MedAustron
- Magnet input parameters, requirements and constraints
- Exercise 1:
  - Analytical design (paper & pencil) to derive the main parameters
  - Expected results: parameter list, magnet cross-section (yoke & coils) ready for entering the model in FE-code
  - Oral presentation: 23rd February 2026 (15’ per group)
  - Please submit your presentations (1 per group) to the JUAS secretariat by email before 23rd February 2026 @ 8:30
- Exercise 2:
  - Numerical field computations and optimization of the pole profile
  - Expected results: written design report with magnet cross-section and optimized pole profile
  - Cost estimate with detailed cost breakdown
  - Please submit your design report (1 per group) to the JUAS secretariat by email before 2nd March 2026 @ 12:00

Introduction to MedAustron

MedAustron is located in Wiener Neustadt (50 km south of Vienna) next to the future site of the new hospital

- Medical Treatment and Non-clinical Research
- Synchrotron based (circumference 76 m)
- Ion species: protons and carbon ions
- Energy range:
  - Proton: 60-220 MeV (medical)
  - Carbon: 120-400 MeV/n



ebg MedAustron

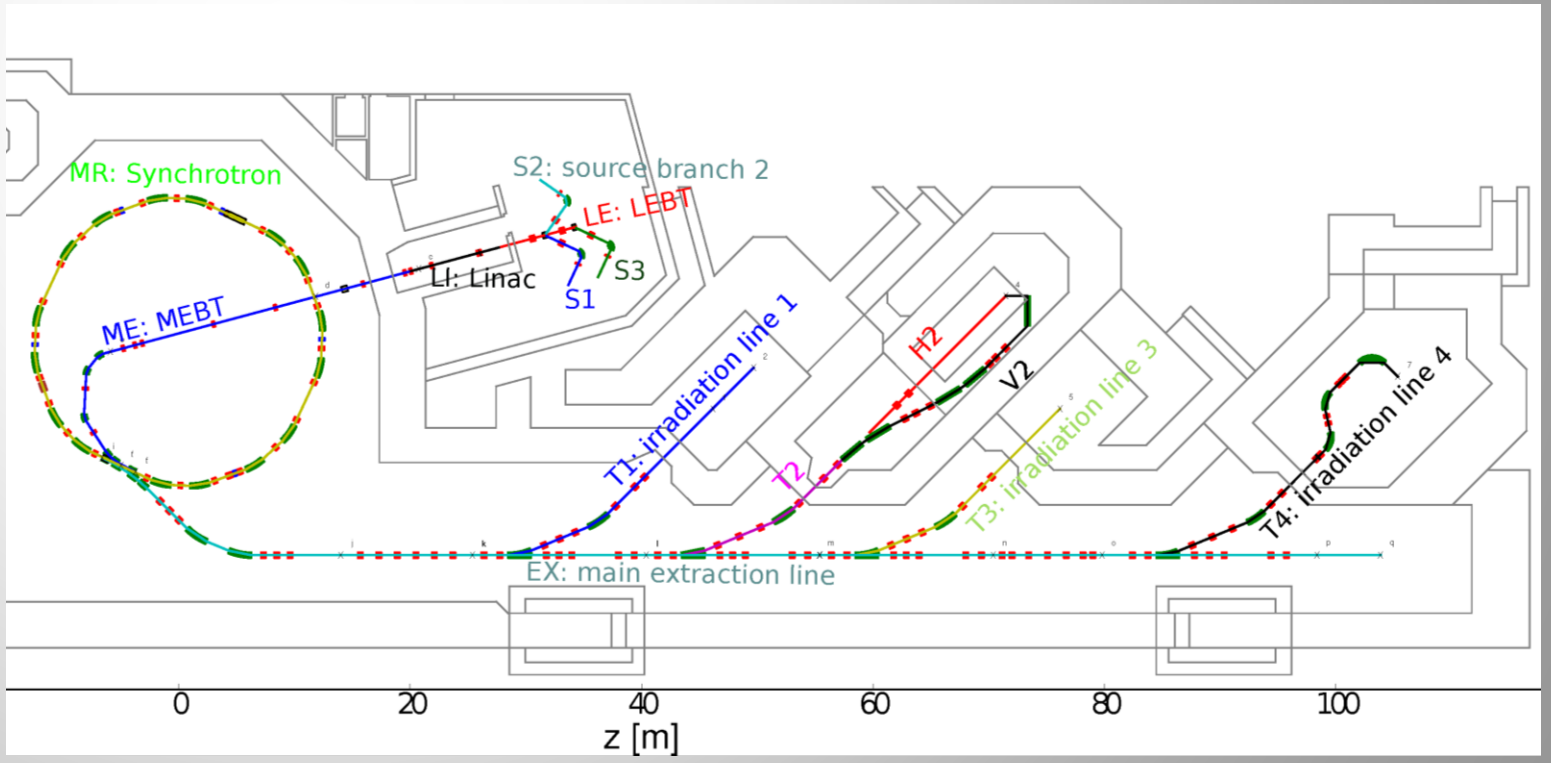
Irradiation rooms

Medical facility:

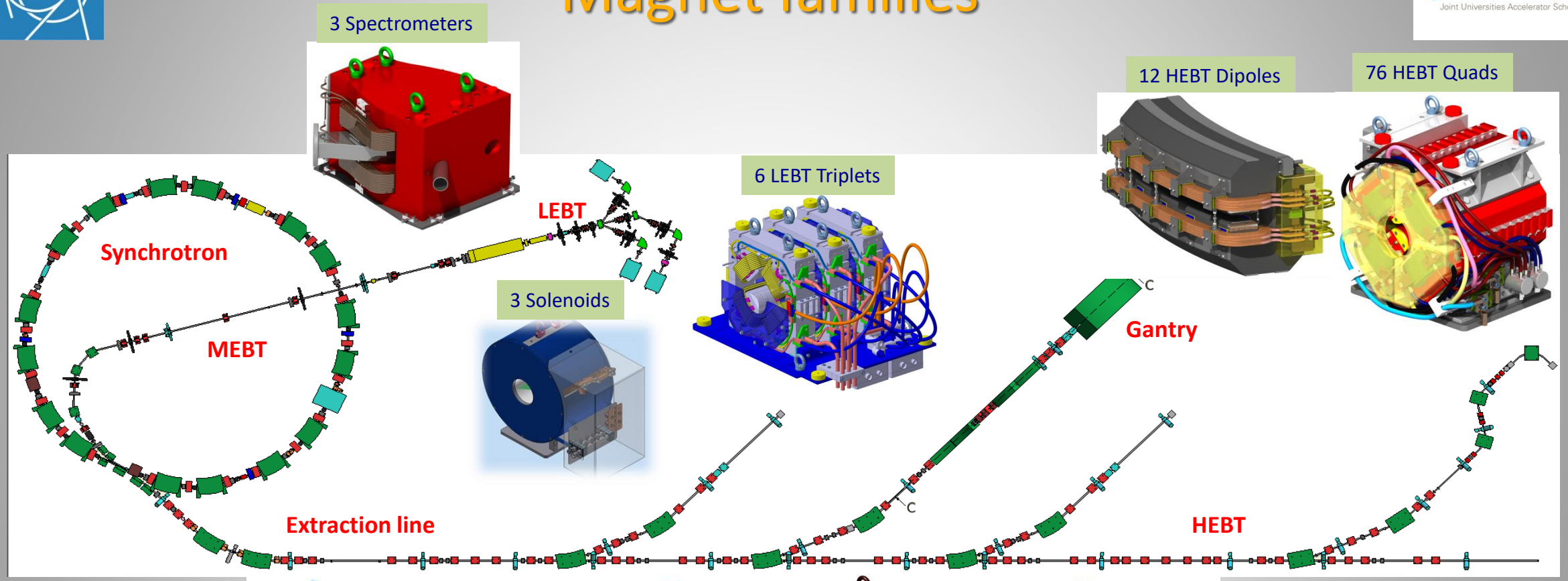
- IR2
  - Horizontal and vertical beam
  - Protons and carbon ions
- IR3
  - Horizontal beam
  - Protons and carbon ions
- IR4
  - Gantry
  - Protons

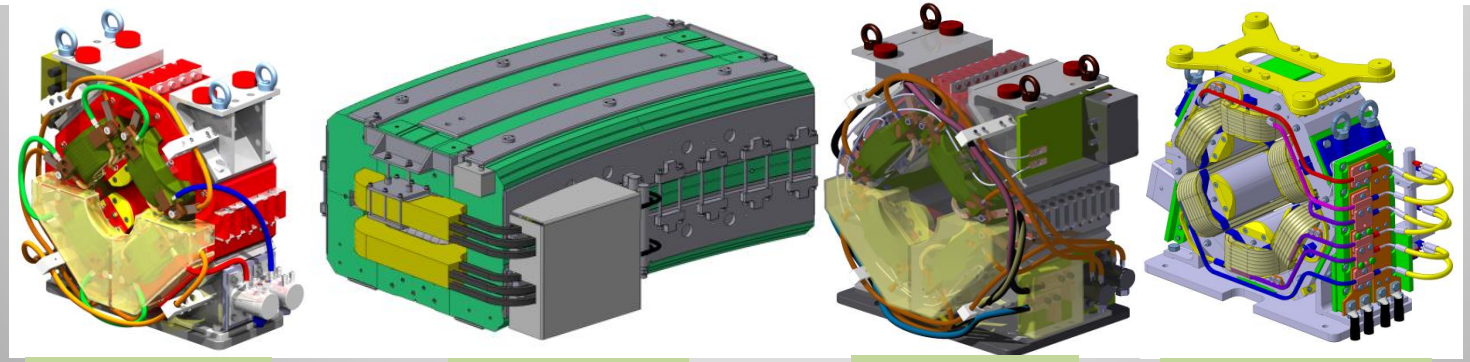
Non-clinical research facility:

- IR1
  - Horizontal beam line
  - Protons (up to 800 MeV) and carbon ions



Magnet families





3 MEBT Dipoles

10 MEBT Quads

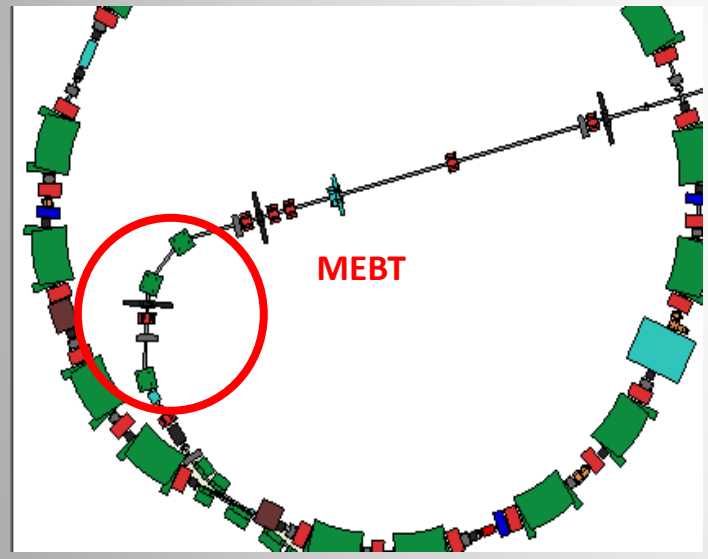
16 Ring Dipoles

24 Ring Quads

5 Ring Sextupoles

Beam parameters

Required: Three bending magnets for the medium-energy beam transfer line (MEBT) between the Linac and the Synchrotron



Parameter	Value	Unit
Particle type	Protons, C^{6+}	
Beam energy	7	MeV/u
Length of beam line	40.9	m
Beta function β_x, β_y	≈10	m
Beam size σ_x, σ_y	± 10	mm

Functional specification

Parameter	Value	Unit
Number of magnets (+ spares)	3 (+1)	
Bending angle (per magnet)	36	deg
Max. mechanical length per magnet l_{mech}	0.9	m
Horizontal good-field region	± 30	mm
Vertical good-field region	± 23	mm
Field quality inside GFR $\Delta B/B_0$	$< \pm 1 \cdot 10^{-3}$	
Available cooling water pressure P_{in}	10	bar
Required return pressure P_{out}	2	bar
Inlet water temperature T_{in}	20	°C
Max. temperature increase on magnet ΔT	15	°C
Max. converter current	600	A
Max. converter voltage (3 magnets in series)	100	V

Other constraints

These additional constraints need to be respected:

- The magnet is operated in quasi-static mode
- However, the magnet yokes shall be built from laminated steel already on stock (type M1200-100A)
- Only one power converter is available: the three magnets have to be electrically connected in series
- One complete spare unit will be required
- If possible, one of the following copper conductors (on stock) shall be used:

Conductor type	*Copper grade*	*Outer dimensions*	*Diameter of cooling duct*	*Edge rounding radius*	*Available length*
Type 1	OF-Cu	25 mm x 25 mm	12 mm	1.5 mm	≈ 300 m
Type 2	OF-Cu	11 mm x 11 mm	5.5 mm	1 mm	≈ 600 m
Type 3	OF-Cu	11 mm x 11 mm	5.0 mm	1 mm	≈ 800 m
Type 4	OF-Cu	8 mm x 8 mm	5.0 mm	1 mm	≈ 500 m

Exercise 1: Analytical design

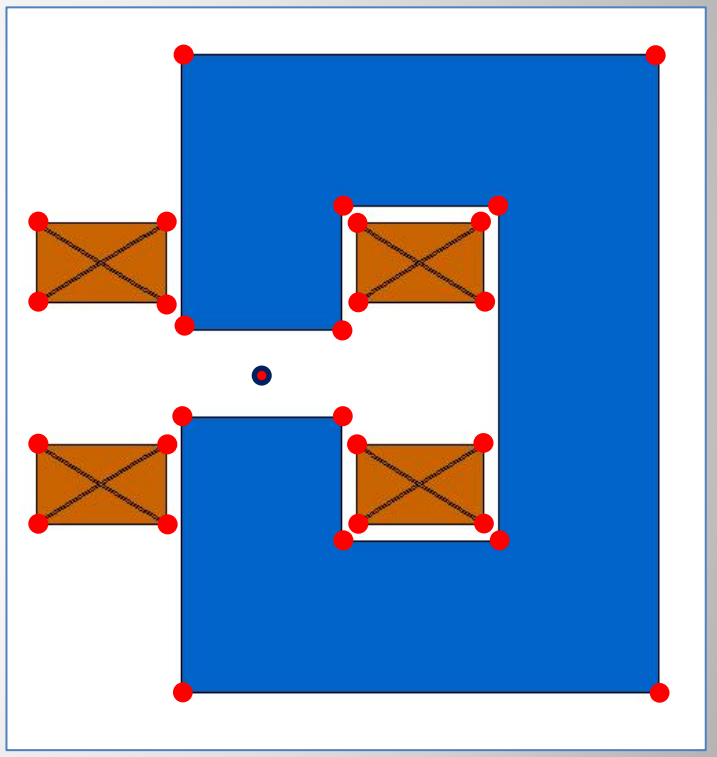
The following parameters should be defined or derived:

- ❑ Magnet type: straight or curved?
- ❑ Magnet shape: H-shape or C-shape?
- ❑ Aperture height h
- ❑ Yoke length l_{yoke} and effective length l_{eff}
- ❑ Flux density B
- ❑ Pole width w and return yoke thickness
- ❑ Excitation current NI (ampere-turns)
- ❑ Coil cooling: water cooling or air cooling?
- ❑ Nominal current I and number of turns N
- ❑ Conductor dimensions and current density J
- ❑ Coil size: coil width, coil height, insulation thickness, conductor length
- ❑ Coil resistance R, voltage drop U and dissipated power P_{Ω}
- ❑ Coolant flowrate Q and required pressure drop ΔP
- ❑ Average coolant velocity v_{avg} and Reynolds number Re

Exercise 1: Analytical design

For the computer work with FEMM, you will need the following magnet parameters:

- ❑ Aperture height
- ❑ Pole width (as small as possible)
- ❑ Yoke dimensions (horizontal and vertical)
- ❑ Coil window width and height
- ❑ Coil dimensions (width and height)
- ❑ Coil position
- ❑ Coil excitation (ampere-turns)



Hint: prepare a sketch identifying the key-point coordinates (x/y) wrt the beam axis

Important guidelines and hints

- Review carefully all input parameters before starting your design work (some parameters might be superfluous)
- The magnet can be of C-type or H-type – up to you to decide
- Effective length $l_{eff} \neq$ yoke length l_{yoke}
- For a straight magnet do not forget to take the sagitta into account
- The pole should be optimized – use the correct approach to determine the pole width
- For the cooling circuit design, a series of educated guesses and assumptions are required
- A rectangular coil cross-section gives better results than a square one
- Do not forget to verify your initial assumption before completing your design
- All parameters need to be checked for the entire operation range – from B_{min} to B_{max} – also then in the numerical design

Exercise 2: Numerical design

2D Numerical calculations with FEMM:

- FEMM: 2D FE code for magnetics, electrostatic, heat flow and current flow problems with graphical pre- and post-processors
- Licensed under the terms of the Aladdin Free Public License
- Input via GUI or scripts (LUA or Octave scripting engine)
- More info (wiki) and download from the web:

http://www.femm.info/wiki/HomePage

Exercise 2: Numerical design

The goal of this exercise is:

- Download, install and get familiar with FEMM
  - Attention: FEMM runs only under WINDOWS
  - MAC users: please use a WINDOWS emulator (e.g., WINE)
- Enter a simple dipole geometry (yoke + coils) using nodes, segments and block labels
- Define and set the necessary material parameters
- Apply the correct boundary conditions
- Analyse the results:

  field lines, flux distribution, central field, field along the axis and the GFR, current density, etc...
- Adjust the pole profile to optimize and reach the required field quality

Cost estimate

Establish a cost estimate for your final magnet design

The cost estimate shall include:

- Investment cost for the 3 magnets + 1 spare unit
- Operation costs (electricity) for the magnets over a lifetime of 20 years
- 70% proton operation and 30% carbon ion operation

Following costs factors shall be considered:

Design work: 15 k€
Stamping tool: 15 k€
Stacking tool: 12 k€
Winding tool: 10 k€
Impregnation mould: 18 k€

Steel: 1.8 €/kg
Copper conductor: 18 €/kg
Yoke manufacturing: 12 €/kg
Coil manufacturing: 36 €/kg
Auxiliary parts: 5 k€/magnet

Electricity: 95 €/MWh
Operation hours: 4200 hrs/year

Please prepare a cost sheet with a detailed cost breakdown

Design report

Students are expected to deliver a written magnet design report which should include at least:

- ❑ Detailed magnet parameter list summarizing the outcome of the analytical design
- ❑ Explanations for your design choices
- ❑ Magnet cross-section based on analytical calculations with yoke and coil shape
- ❑ Optimized pole profile based on numerical computations, fulfilling the field quality requirements
- ❑ Results from the numerical simulation (field quality plot) for the entire operation range
- ❑ Cost estimate with detailed cost breakdown, including investment and operation costs
- ❑ Important: the reports will be evaluated and contribute to the total score together with the results of the exam
- ❑ Please send the final report (1 per group) in electronic form, plus the FEMM input file (*.FEM) of your final FE-model to the JUAS secretary
- ❑ Submission deadline: Monday, 2nd March 2026 @ 12:00 (sharp!)

Good luck...

... and have fun!

... and it is not forbidden to ask questions!